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PHAR0002: An introduction to the mechanisms of drug action.

More therapeutic targets: G protein-coupled receptors and receptor tyrosine kinases.

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Aims:

1) To understand the molecular basis of G protein-coupled receptor (GPCR) signalling.
2) To understand the molecular basis of receptor tyrosine kinase (RTK) signalling.
3) To appreciate the relevance of GPCR and RTK signalling to cholera, anaphylactic shock, asthma and diabetes mellitus.
4) To be able to discuss named examples of drugs targeting GPCR and RTKs receptors.

G protein-coupled receptors as drug targets.

G protein-coupled receptors (GPCRs, also known as 7TM receptors) are proteins located in the cell membrane that detect signals from the outside world and produce a response in the cell interior. GPCRs form one of the largest families of proteins found in the human body. As you might therefore expect, they perform a very diverse set of physiological functions. For example, G protein-coupled receptors are needed to detect photons in your eye and to detect odours in your nose. In addition, your body uses G protein-coupled receptors to speed up (and slow down) the heart, to regulate blood pressure, to regulate the secretion of stomach acid and many more things besides. Because of the importance of their physiological roles and because of the diversity of their function, it should not come as a surprise to discover that many different drugs are targeted to act on G protein-coupled receptors. These include sumatriptan (drug company name Imigran) for the treatment of migraine, cetirizine (drug company name Zirtek) a commonly used antihistamine, salbutamol (drug company name Ventolin) for the treatment of asthma and atenolol (drug company name Tenormin) for the treatment of hypertension (high blood pressure). In fact, a few years ago it was estimated that approximately ~30% of top selling drugs were targeted towards this type of receptor.

How these receptors work.

Fortunately, despite their diversity of function in the human body, all G protein-coupled receptors operate, at a molecular level, in a very similar way. In a moment we will describe the principal steps involved in GPCR signalling but before we do this, two things must be explained. First, the meaning of the term *agonist*, second what is a G protein.

What is an agonist?

An *agonist* is a substance that interacts with a receptor so as to activate it. Agonists form the signals that GPCRs detect. Thus, in most cases, G protein-coupled receptors and receptor tyrosine kinases are inactive until they are activated by the binding of *agonist* molecules.

What is a G protein?

G proteins themselves are proteins that bind the **guanine** nucleotides, guanosine triphosphate (GTP) and guanosine diphosphate (GDP); this is where the 'G' in 'G protein' comes from. There are a number of different 'super families' of G proteins (families which themselves contain families of proteins). The type that associates with G protein-coupled receptors is called a heterotrimeric G protein. Heterotrimeric G proteins consist of three different subunits (hence the constituents of the name; 'hetero' meaning different and 'tri' meaning three). First, there is the alpha subunit (Gα). This is the one that actually binds a GTP or GDP molecule. Then there are two additional subunits; the beta (Gβ) and gamma (Gγ) subunits. The Gβ and Gγ subunits stick together very tightly and always remain attached to the cell membrane. Because of their close association they are often written as if they were a single entity (G$\beta\gamma$) which, for all practical purposes, they are. The Gα and the Gγ subunits both have lipid molecules attached, tethering them to the cell membrane. In the resting state all three G protein subunits are assembled together as a G$\alpha\beta\gamma$ complex, which has a molecule of GDP bound to the alpha subunit.

Now that we know something about G proteins we can discuss how they couple to G protein-coupled receptors.

Receptor G protein coupling.

In the human body, at least some of the G protein-coupled receptors operate as dimers. However, it seems likely that in most cases only a single receptor protein is needed to couple to the G protein. Hence, for the sake of simplicity, that is how we will consider them here. The basic steps in coupling between G proteins and G protein-coupled receptors are illustrated in Figure 1. These steps are:

Step 1.

Things start with the receptor sitting in the membrane in an inactive state. Nearby, or perhaps already bound to the receptor there will be an inactive G protein (a G$\alpha\beta\gamma$ complex with GDP bound as described above).

Step 2.

Agonist binding to the receptor changes the receptor's conformational state. It now interacts with the G protein in such a way that it enables the G protein to undergo its own conformation change, releasing the bound GDP molecule and binding in its place a molecule of GTP.

Step 3.

The binding of GTP enables the G protein alpha subunit to undergo a further conformation change and separate from its beta and gamma partners. Although the beta and gamma subunits remain bound together, they too leave the receptor. In this separated form, the G protein alpha subunit and/or the G$\beta\gamma$ subunits 'transduces' a physiological response inside the cell by activating or inhibiting '*effector*' proteins (we'll discuss these in a moment).

Step 4.

The activation of effector molecules lasts until the G protein alpha subunit catalyses the conversion of GTP to GDP. At this point, when the Gα has GDP bound, it leaves the effector and rejoins its beta and gamma partners, ready for the cycle to start again.

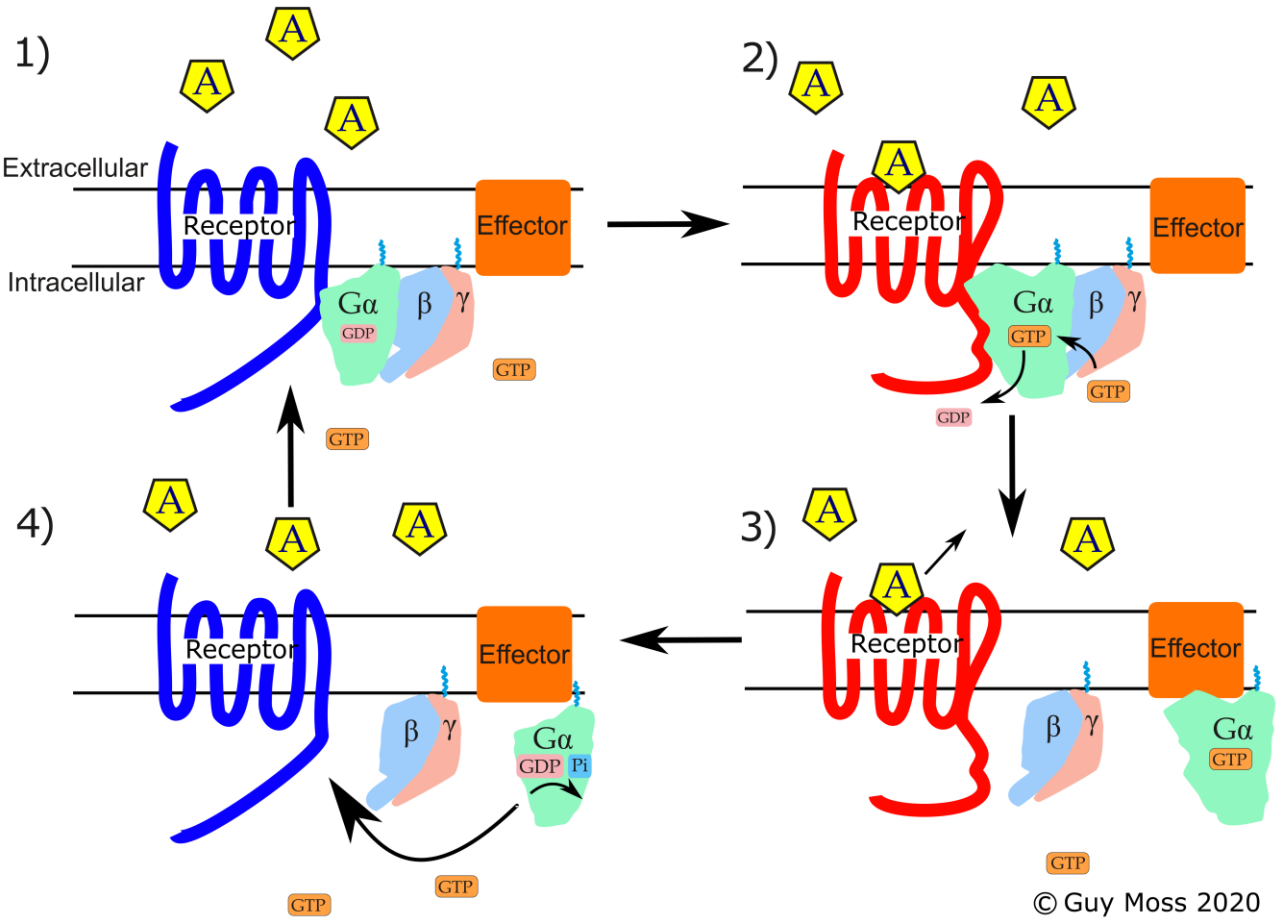


Figure 1. The G protein-coupled receptor cycle. 1) Initially, the receptor has no agonist bound, is inactive, and sits in the membrane with an equally inactive G protein. 2) Agonist binding causes the receptor to undergo a conformational change and this new receptor conformation allows the G protein to undergo its own conformational change and release GDP, replacing it with GTP. 3) Once GTP is bound the Gα and Gβγ subunits dissociate and diffuse away from the receptor to interact with effector molecules, bringing about the physiological response. 4) The cycle ends when the Gα subunit breaks down GTP, leaving a GDP molecule bound. In this form it separates from the effector and rejoins the Gβγ subunits, ready to restart the cycle.

What are 'effectors'?

Effectors are either cell enzymes or ion channels that are targets for G protein modulation. One example is adenylyl cyclase, a protein that produces cyclic 3',5'-adenosine monophosphate (cAMP). The binding of certain types of Gα can stimulate this enzyme which then increases cellular levels of cAMP. In turn, raised

cAMP levels activate cAMP-dependent protein kinase A (PKA) and consequently produce phosphorylation-dependent regulation of other proteins such as ion channels.

G protein-effector interactions.

There are a number of different types of G proteins that interact with different effectors. G proteins are usually referred to by the subtype of alpha subunit that they contain. Perhaps the three most well-studied families are the G_s, $G_{i/o}$ and G_q G proteins and because of the naming convention these carry the $G_{\alpha s}$, $G_{\alpha i/o}$ and $G_{\alpha q}$ subunits respectively. The G_s family of G proteins *stimulate* the membrane protein adenylyl cyclase, as discussed above. The $G_{i/o}$ family can *inhibit* adenylyl cyclase and interact with a variety of targets including ion channels. The G_q family can activate the enzyme phospholipase C. This enzyme breaks down the cell phospholipid PIP_2 into diacyglycerol (DAG) and inositol (1,4,5)-trisphosphate (InsP3 or IP3). In turn, this $InsP_3$ releases calcium from intracellular stores, and thus activates further processes. The coupling between specific G proteins and GPCRs determines the cell's physiological response, as shown in Figure 2.

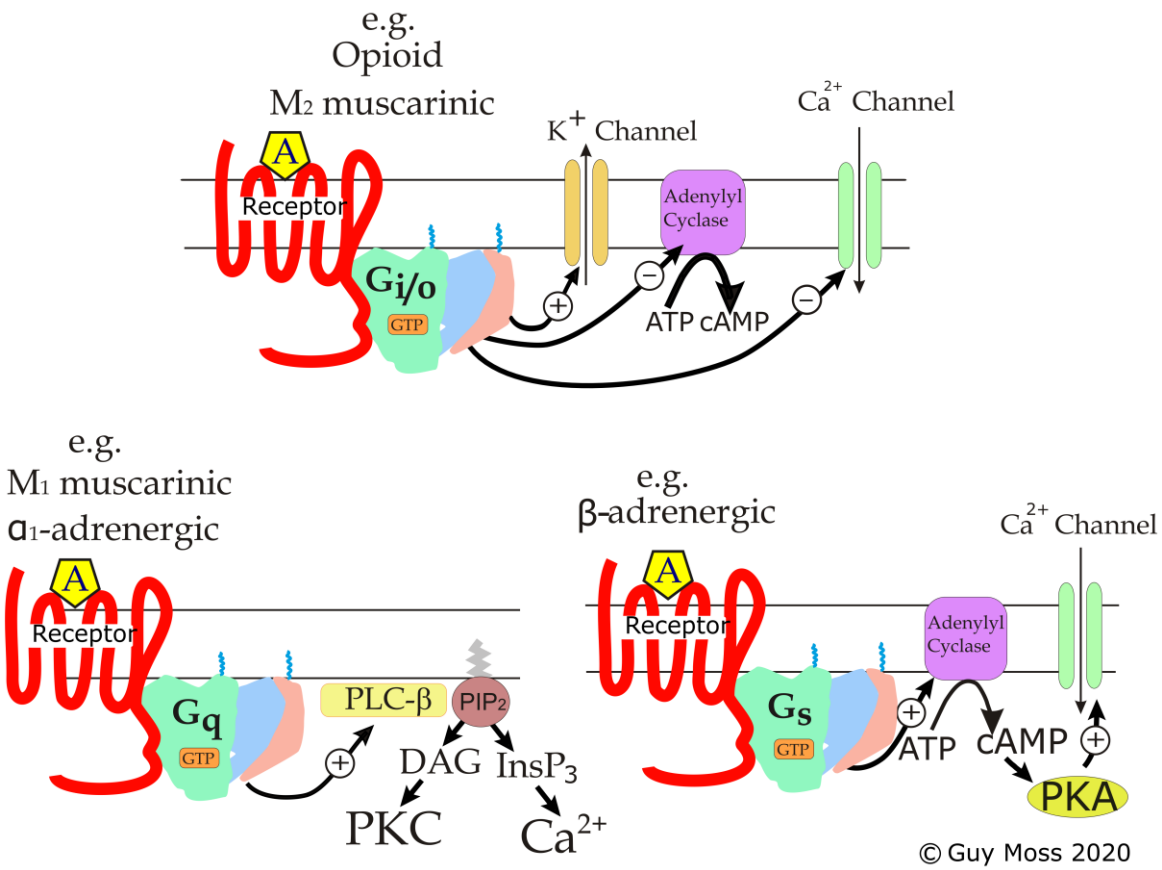


Figure 2: Examples of G protein receptor coupling to target enzymes and ion channels. Opioid and acetylcholine muscarinic M2 receptors couple to $G_{i/o}$ subunits and can thus down-regulate adenylyl cyclase or affect ion channel activity. Acetylcholine muscarinic M1-type receptors and α1-type adrenergic receptors, on the other hand, couple to G_q G proteins and thus can stimulate phospholipase C (PLC-β) which, in turn, breaks down the phospholipid PIP2, creating diacyglycerol (DAG) and InsP3. DAG can then activate protein kinase C (PKC), while InsP3 can release Ca^{2+} from intracellular stores. Finally, receptors such as the β–adrenergic receptor couple to G_s and thus activation of these receptors stimulates adenylyl cyclase producing cAMP, which activates proteins kinase A (PKA). Note: because agonists like acetylcholine can couple to multiple receptor subtypes, the same signaling molecule can produce different physiological responses in different cell types.

G proteins and cholera. The powerful role of these G proteins in our body is well illustrated by cholera toxin. In a case of cholera toxin poisoning, the toxin is secreted by invading bacteria and is subsequently able to enter cells in the body, catalyzing the ADP ribosylation of some subtypes of G protein alpha subunits. This ribosylation has the effect of leaving the G protein permanently "switched on". The main symptom of cholera toxin poisoning, the excessive loss of fluid from the gastrointestinal track (causing diarrhoea), is largely accounted for by this G protein activation. The resultant loss of fluids and electrolyte is the main, immediate threat to life. Take-home message – over-activity of your G proteins can be fatal!

Some examples of G protein receptor pharmacology.

Adrenaline in the treatment of anaphylactic shock.

A good example of the power of drugs working through G protein-coupled receptors can be found in the treatment of anaphylactic shock. This is an allergic over-reaction of the body that can be set off by a number of different stimuli in susceptible individuals. For example, some people can have very severe responses to bee stings, others have an adverse reaction to drugs such as aspirin and some people react badly to certain foods such as fish, nuts and cow's milk. Whatever the cause, the immediate symptoms of anaphylactic shock are laryngeal oedema (swelling of the larynx), bronchospasm (constriction of the airways) and hypotension (a loss of blood pressure). In this case, the first line treatment is to inject adrenaline (epinephrine) which, acting through G protein-coupled receptors, reverses these immediate symptoms. After a moment of thought it shouldn't come as a surprise that adrenaline is able to do this. After all, the natural release of adrenaline in the body is associated with exercise, excitement and activity. These are circumstances in which you might need an increased blood pressure, dilated airways for deeper breathing and an increased cardiac output (which will also tend to increase blood pressure). Because this is a life-threatening condition, people known to be susceptible to this problem sometimes carry 'EpiPens®', to self administer adrenaline, in the event of an emergency.

Adrenaline-like agonists in the treatment of asthma.

We have just mentioned that one effect of adrenaline is to reverse bronchospasm in cases of anaphylactic shock, a problem in which there is a widespread overreaction of the immune system. In asthma, the problem is more localized in that an adverse immune reaction is triggered in the lungs causing constriction of the airways. However, the way in which this problem is tackled is often somewhat similar because one of the first line treatments for an asthmatic attack is to inhale an adrenoceptor agonist, such as salbutamol which, like adrenaline, activates adrenoceptors in the lung smooth-muscle, relaxing them and causing bronchodilation. The advantage of taking this drug by inhalation is that it is applied where it is needed - directly onto the lungs. Further, the dose becomes diluted by the time it reaches other targets where adrenaline-like effects are not wanted (such as the heart). Even so, tremors and cardiac effects will occur. Another way in which pharmacologists have tried to minimize these unwanted side-effects in the treatment of asthma is by using drugs that are designed to preferentially interact with the subtype of adrenoceptor found in lung smooth-muscle ($\beta 2$) over the subtype that causes most cardiac effects ($\beta 1$).

Receptor tyrosine kinases.

Receptor tyrosine kinases often mediate the signals of peptides that regulate cell growth and division. Consequently, there is much interest in them as possible therapeutic targets particularly for anti-cancer drugs (since cancer is a problem of uncontrolled cell growth). One such example is the drug Erlotinib that inhibits epidermal growth factor receptors (EGFR). It is currently recommended by NICE (National Institute

for Health and Clinical Excellence) as an option for treating non-small cell lung cancer that has progressed. Unfortunately, the impact of Erlotinib is often quite modest.

The structure of receptor tyrosine kinases (RTKs) can be broadly divided into three regions; the extracellular region where ligand binding occurs, a transmembrane region which connects to the cell interior, and an intracellular portion that contains the tyrosine kinase element which gives them their name. In the resting (inactive) state, most RTKs sit in the membrane as monomers i.e., a single protein molecule on its own. There are some exceptions to this however, including the insulin receptor which is composed of two short alpha chains disulphide-linked to two transmembrane beta chains. The alpha chains are also linked to one another by disulphides, thus forming an $\alpha 2\beta 2$ heterotetramer as shown in Figure 3.

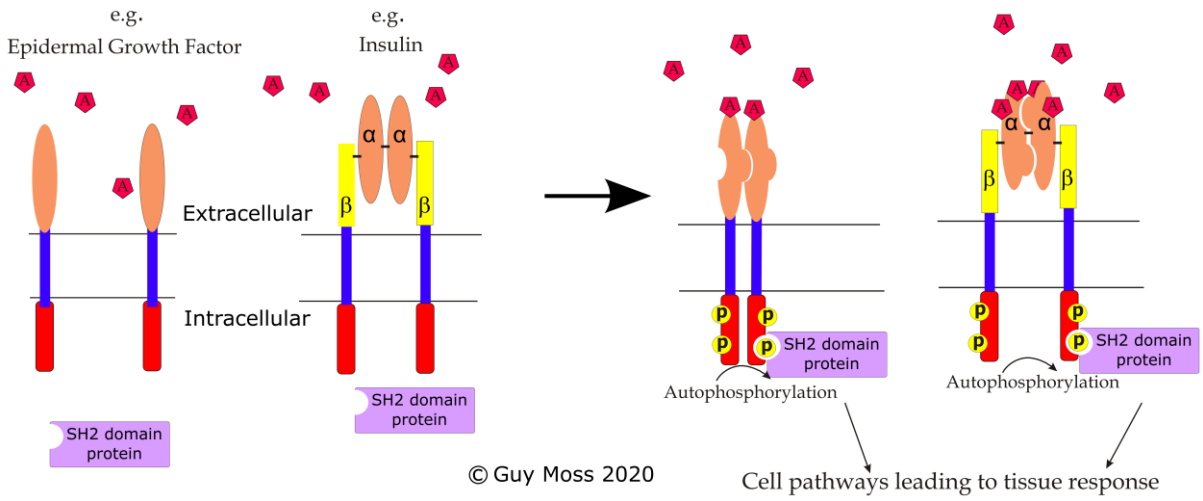


Figure 3. The principal steps in receptor tyrosine kinase (RTK) signalling. In the absence of agonist, most RTK receptors sit in the membrane as monomers, although a few, such as the insulin receptor, are present as inactive dimers (actually, a dimer of dimers as described above). On agonist binding, the receptors form an active receptor dimer, sometimes via a conformational change and sometimes via the agonist binding two receptor molecules. In this active state the receptor phosphorylates itself (autophosphorylates) on specific, intracellular tyrosine amino acid residues. The phosphorylated portions of the RTK receptor are then recognized by other proteins that contain special recognition domains, such as the SH2 domains shown here. The binding of these SH2 domain proteins to phosphorylated RTK receptors, causes activation of the SH2-domain protein. In turn, this leads to further steps in diverse and sometimes complicated signaling pathways that produce the cellular response.

Receptor tyrosine kinase signalling.

1) In the absence of agonist, receptor tyrosine kinases sit in the membrane in an inactive state, usually as monomers (although some receptors, such as the insulin receptor discussed above, form an inactive dimer, or more exactly in the case of insulin, a dimer of dimers).

2) For monomeric receptor tyrosine kinases agonist binding can cause dimerization in a number of ways. For example, one agonist molecule may be capable of binding two such monomers at once. When an agonist does this, it can simply act as a 'bridge' to cross-link the receptor monomers and bring about activation. Alternatively, the agonist may cause a conformational change in the extracellular region that promotes the formation of receptor dimers. Whichever way it happens, once dimers are formed, changes in the intracellular regions of the receptor occur that activate their kinase domains. (How this step happens

is still being actively researched but it seems likely that in many cases a cis-auto-inhibition (self-inhibition) system is released and the kinases, which previously had a very low level of activity, suddenly become fully active.)

3) Once fully active trans-autophosphorylation of the RTK kinase domains occurs (i.e. phosphorylation by the opposite partner in the kinase dimer). This creates phosphotyrosines, phosphorylated tyrosine residues that act as specific binding sites for the assembly of downstream signaling molecules. These downstream proteins contain special structural domains that interact with the kinase domain. Two such binding domains are the phosphotyrosine binding domain (PTB domain) and the Src- homology 2 domain, or SH2 domain, which gets its name from the protein 'Src', where this type of structure was first identified.

4) Binding of a protein via an SH2 or PTB domain activates further steps in the signalling pathway and may lead to binding of additional proteins to the signalling complex. The precise nature of the signalling elements varies enormously. However, as an example, some RTKs activate phospholipase C-γ, which binds via its SH2 domain. It is then targeted to the plasma membrane where it causes the breakdown the lipid PIP2 and the formation of IP3 and diacylglycerol in a manner similar to that seen with some G protein-coupled receptors.

<u>Insulin receptors and diabetes.</u>

Insulin is a hormone that controls blood glucose levels. This is necessary because, for example, following a meal there is more glucose available than needed and the excess fuel must be stored. On the other hand, between meals we need to call on those stored energy reserves.

Diabetes mellitus is a form of diabetes characterised by hyperglycaemia (high blood glucose levels) and this is an indication that the body's fuel is not being stored properly. Blood is filtered to form urine and the excess glucose, by osmosis, takes water with it, increasing the volume of urine. Without the necessary supply of energy, the body starts to breakdown triglycerides (fats). The products of this breakdown are then metabolized to acetoacetate and then to β-hydroxybutyrate and acetone. Acetoacetate and β-hydroxybutyrate are highly acidic 'ketone bodies' and diabetic ketoacidosis occurs when the blood becomes more acidic than body tissues. This is very damaging and can be fatal if it continues unchecked. Diabetes mellitus can have two forms, Type I and Type II. Type I usually begins in childhood (juvenile onset diabetes) and is caused by a severe under-production of the hormone insulin, often following on from destruction of the body's insulin-producing cells. This type of diabetes can only be treated with insulin and without treatment patients will die. Type I diabetes is thus called insulin-dependent diabetes mellitus (IDDM). Type II diabetes mellitus usually has a late onset (maturity onset diabetes) and is characterised by both an impaired response to insulin in the body and impaired insulin secretion. Because weight loss and diet control can be effective treatments for this condition, it is sometimes called non-insulin dependent diabetes mellitus (NIDDM). This is perhaps a slightly misleading name, since when other treatments fail, patients with NIDDM will probably be prescribed insulin.